

best\_india.pt — 300 buildings in densest tile





Q.1 Satellite images are huge files — hundreds of megabytes or even gigabytes in size. Downloading live means we only fetch the exact area and zoom we need, making it fast and practical to share.

Q.2 (a) No. of tiles increase  $\sim 4$  times (double in both  $x$  and  $y$ )

(b) Rooftops become clearer —  $0.3\text{ m/pixel}$  vs  $0.6\text{ m/pixel}$ , so small irregular Indian rooftops are easier to detect.

Q.3

- Lowering to  $0.4 \rightarrow$  more buildings detected but risk counting fake ones (trees, shadows)
- Raising to  $0.9 \rightarrow$  fewer buildings detected but risk missing real ones.
- $0.65$  balances both errors reasonably

Q.5 Sandbox weights already learned general features like edges, shapes, textures. Fine tuning only teaches the difference - what Indian rooftops look like. So you need less data and fewer training steps.

Q.6 mAP checks if your box matches the label's exact shape and position. Open Buildings labels are themselves imperfect in dense areas. So our box can visually sit on a rooftop perfectly but still fail mAP because it doesn't match the exact label polygon.

Q.7 A 20-floor tower and 1-storey hut both count as "1 building." The tower houses 200 families, the ~~house~~ hut houses one. Also some buildings are like used for commercial purpose or some are still under construction or abandoned.



Q.8 Not a waste. Open buildings doesn't exist everywhere or for every time period.

~~Building our own model~~

Building our own model taught us domain gap, confidence thresholds, and precision vs recall-concepts, <sup>we</sup> ~~can't~~ can't learn just by querying a database.

Code Part Questions:-

Q.1 The sandbox model trained on US/  
European cities (specifically on <sup>a city like</sup> Paris)

where buildings are large, rectangular and well separated. Lucknow's buildings are small, irregular and packed together.

~~High recall in session 1 only means it was good on that specific data.~~

Q.2 Total buildings detected by our model is 4638.

There are some shun areas where rooftops are packed so tightly, the model can't separate individual buildings there so there is a chance where it can miss. ~~some~~

Q.3 Open buildings was trained on far more data with more compute, specifically fine-tuned for the Global South including India.

~~we should use~~ we should use existing dataset when it's more accurate than what we can build in our limited time and budget. We should build our own only when no dataset covers our region, time period, or needed features.